

KIS Documentaiton

System Documentation - Hospital Information System (KIS)

This document provides a detailed overview of the main functions and capabilities of the Hospital Information System (KIS) application.

1. Overview

The KIS application is a comprehensive software solution designed to streamline hospital operations, including patient admission, case management, ward monitoring, and communication with external medical systems via HL7. The application features a robust JavaFX-based graphical user interface (GUI) with a fixed-size layout (1030x600) to ensure a consistent user experience across different workstations.

2. Core Main Functions

2.1 Secure Authentication & Session Management

- ***Login System****: Access is restricted to authorized personnel. Users must select their employee profile from a pre-loaded list to log in.
- ***Role-Based Access****: Upon logging in, the system grants access to functional tabs (Admission, Overview, Communication).
- ***Session Tracking****: The application maintains a global session (`AppSession`) tracking the currently logged-in user, which is required for auditing actions like creating new patient cases.
- ***Logout****: Users can securely log out, which disables access to sensitive data and resets the interface.

2.2 Patient Admission (Aufnahme)

The Admission module is the entry point for patient processing.

- ***Patient Search****: Features an intelligent auto-complete search field that allows staff to quickly find patients by last name.
- ***New Patient Creation****: A dedicated window (`PatientController`) enables the registration of new patients with validation for mandatory fields.
- ***Patient Editing****: Existing patient records can be updated via the "Edit Patient" dialog.
- ***Case Management (Fallaufnahme)****:
 - * ***Case Creation****: Staff can create new medical cases for a selected patient.
 - * ***Doctor & Ward Selection****: Assigning an attending physician and an initial ward/station.
 - * ****nsurance (Kostenträger)****: Supports multiple insurance providers per case, with built-in validation to ensure the coverage allocation totals exactly 100%.
 - * ***Validation****: Prevents the creation of a new case if the patient already has an active, undischarged case.

2.3 Clinical Overview (Übersicht)

The Overview tab serves as the central dashboard for monitoring hospital activity.

- ***Patient List****: Displays a list of currently admitted patients.
- ***Ward Overview (Stationsübersicht)****: Provides a view of patient distribution across different wards.
- ***Case & Operation Status****: Monitors the status of ongoing cases and scheduled surgeries.

- ***Global Refresh****: A synchronized refresh mechanism ensures that all lists and views are updated in real-time across the application when data changes (e.g., after a new admission).

2.4 Communication Interface (HL7)

The application includes a standard HL7 (Health Level 7) interface for interoperability with other medical systems (e.g., Laboratory, Radiology).

- ***Message Generation****: Ability to send patient and case data as HL7 messages.
- ***Network Configuration****: Configurable IP address and Port (default: localhost:39069) for the destination system.
- ***Message History****: A log view displays sent messages with their status and timestamps, ensuring traceability of data exchange.

2.5 Infrastructure & Utilities

- ***Database Integration****: Uses H2 database via customized Repositories for persistent storage of patients, cases, and employee data.
- ***Logging****: comprehensive logging (SLF4J) for debugging and audit trails.
- ***Global Layout Policy****: Enforces a strict non-resizable window policy to maintain UI integrity.

3. Technical Architecture

- ***Framework****: JavaFX 17+
- ***Architecture Pattern****: MVC (Model-View-Controller) with Dependency Injection.
- ***Build Tool****: Maven
- ***Database****: H2 SQL Database